

PC NF-NAI

Energy-Efficient Traffic Steering for 5G and Beyond Slicing



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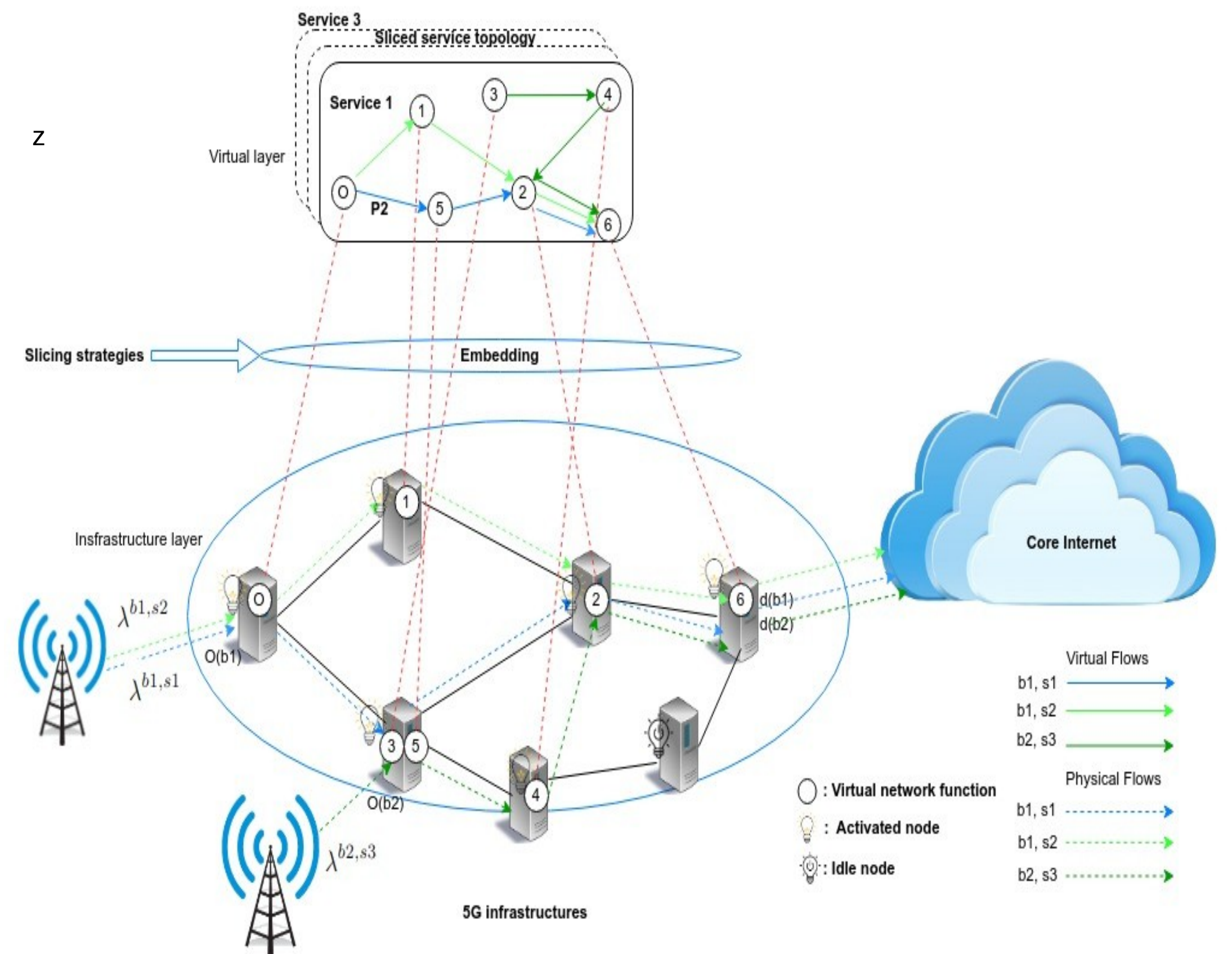
BACKGROUND & MOTIVATION

- 3GPP proposes network slicing, a primary features of modern 5G and beyond (5G&B) networks[1].
- Telcos, e.g., Ericsson, advocate for transparent and energy-efficient implementations of 5g&B networks [2].

Given a network topology and a virtualized slice service pool : How can we guarantee both slice performance and low energy consumption ?

We assume that we can increase Efficiency in the network with resource allocation and traffic steering.

- With the aims to improve efficiency in the network, we propose in this work to solve energy-efficient 5G slicing problem by considering resource Overbooking features.
- Compared to other works in the state-of-the-art, where overbooking is used to maximize MNOs' profit (e.g., [3,4]), our work envisions using overbooking to minimize energy consumption.



PROBLEM

We model the Energy-Efficient 5G Slicing Problem within the Stochastic Optimization Framework.

$$\text{Minimize}_{x,y,z} \sum_{i \in V} \varepsilon_i^{\text{activ}} \cdot x_i + \sum_{b \in B} \sum_{s \in S} \varepsilon_{ij}^{\text{comp}} \cdot y_i^{b,s} + \sum_{j \in V: j \neq i} \varepsilon_{ij}^{\text{comm}} \cdot z_{ij}^{b,s} + \overbrace{E(\sigma_{ij})}^{\text{energy penalty}}$$

Subject to:

- Node activation : $y_i^{b,s} \leq x_i \forall i \in V, \forall b \in B, \forall s \in S$
- Link bandwidth : $\sum_{b \in B} \sum_{s \in S} \lambda^{b,s} \cdot z_{ij}^{b,s} \leq C_{ij} + \underbrace{\sigma_{ij}}_{\text{slack variable}} \forall (i,j) \in L$
- Flow conservation : $\sum_{j \in \delta(i)} (z_{ij}^{b,s} - z_{ji}^{b,s}) = 1_{i=o(b)} - 1_{i=d(b)} \forall b \in B, \forall s \in S$
- VNF placement : $\begin{cases} z_{ij}^{b,s} \leq y_i^{b,s} \wedge z_{ij}^{b,s} \leq y_j^{b,s} \\ z_{ij}^{b,s} \geq y_i^{b,s} + y_j^{b,s} - 1 \end{cases} \forall i, j \in V, \forall b \in B, \forall s \in S$
- Slice performance constraint : E2E delay, node computing capacity.

→ **This problem is NP-hard**

METHOD

To solve the problem we perform Two-Stage Stochastic Programming.

- Explanations :
- Arrival throughput $\lambda^{b,s} > 0$ is the random observation.
- Residual bandwidth $\sigma_{ij} = \begin{cases} \lambda^{b,s} - C_{ij} \leq 0, \text{normal situation} \\ \lambda^{b,s} - C_{ij} > 0, \text{Overbooking situation} \end{cases}$
- First stage $\lambda^{b,s} - C_{ij} \leq 0$
→ Average solution covering all inputs.
- Second stage $\lambda^{b,s} - C_{ij} > 0$
→ Actual solution covering specific inputs.
- Metric analysis energy consumption.
- Performance evaluation :
→ Pure slicing vs energy performance slicing.

Work in progress

SCIENTIFIC COOPERATION

- Potential collaboration with PC7 on accurate energy consumption models

REFERENCES

- [1] The 3rd Generation Partnership Project (3GPP), "Study on enhancement of network slicing; Phase 3", Release 18, Version 1.0.0, February 2024.
- [2] Ericsson Mobility Report. [Online]. Available: <https://www.ericsson.com/49da93/assets/local/mobility-report/documents/2020/june2020-ericsson-mobility-report.pdf>.
- [3] Taskou, S. K., Rasti, M., & Nardelli, P. H. (2022, April). Minimizing Energy Consumption for End-to-End Slicing in 5G Wireless Networks and Beyond. In 2022 IEEE Wireless Communications and Networking Conference (WCNC) (pp. 860-865). IEEE.
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